When Should the Agent Speak?
A Survey of Intervention Timing for Always-On AI Assistants

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Abstract

Capable actors have arrived: agents complete multi-step tasks against verifiable goals, and egocentric perception resolves user intent with high accuracy. What remains unmodeled is the decision of *when* to act unprompted. We survey intervention timing for always-on assistants (smart glasses, MR headsets, ambient copilots) around a single decision rule, intervene iff the expected benefit of acting exceeds the expected cost of interrupting, and organize the literature into five layers: signals, decision, action, memory, and evaluation. We reconnect two lineages that currently do not cite each other: the 1999–2017 interruptibility literature, which formalized interruption cost rigorously but had no capable actor, and the 2024–2026 proactive-agent wave, which has actors but rediscovers the cost term only in fragments (false-alarm pricing, cognitive load, social violation, compute) that no work unifies. We argue that evaluation is the gating layer. Until intervention quality is measurable, a reinforcement-learning reward for proactivity cannot be defined. We therefore propose the design of a benchmark for open-world intervention timing with an explicit cost term. Six open problems close the survey.

1 Introduction

For thirty years, the contract between a user and a computer has been reactive: the machine acts when asked. Always-on devices break the contract. A smart-glasses assistant that waits to be asked is a voice recorder with extra steps; the entire premise of ambient, egocentric, mixed-reality computing is that the system sees what you see and helps *before* you formulate the request. Every vendor building for these form factors therefore ships, or intends to ship, an agent that acts unprompted. That forces a decision reactive systems never had to make: *is this the moment?*

This survey is organized around a single claim: that decision is a priced trade-off,

$$\text{intervene} \iff \mathbb{E}[\text{benefit of acting}] - \mathbb{E}[\text{cost of interrupting}] > \theta, \tag{1}$$

and the state of the field in 2026 is that the left side's first term has never been stronger while its second term has mostly vanished from the models. The threshold θ is not a nuisance constant: with both expectations estimated from noisy signals, θ absorbs calibration error and encodes conservatism. It is also personal, since the same intervention is help for one user and noise for another. Where the benefit and cost estimates are trustworthy, $\theta \to 0$; nowhere today are they trustworthy.

Two literatures hold the two halves of Eq. 1, and they do not cite each other. From 1999 to 2017, the interruptibility and attention-management community formalized the cost side with a rigor the field has since lost: interruption cost as an expectation over an inferred attentional state, priced in dollars, forecastable minutes ahead, learnable from cheap sensors better than human observers

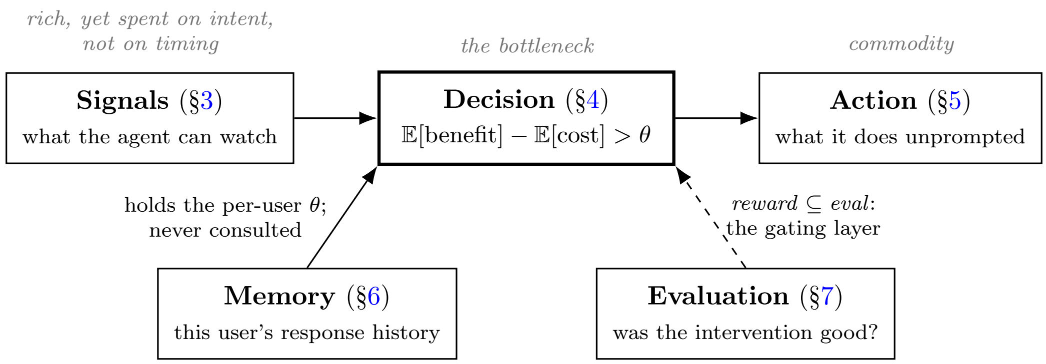

Figure 1: The five-layer taxonomy, annotated with each layer's 2026 health. Perception is rich but aimed at intent rather than timing; the actor is a commodity; the decision layer is the bottleneck. The two lower layers carry the survey's argument: memory holds the user's own response history, the single most valuable timing feature, and no timing policy consults it (§6), while evaluation gates everything, because a training objective for the decision layer can only be written over quantities some evaluation measures (§7).

manage, and ultimately deployed at the scale of ten million users (§2). What that community never had was an actor: the maximum payoff of a perfectly timed decision was a well-placed notification. Since 2024, the proactive-agent community has had the actor, in the form of systems that complete multi-step tasks against verifiable goals (§5), and has rebuilt the decision layer from scratch without the inheritance: rules that fire on script deviation with no cost term, learned relevance models whose false-alarm rates exceed 50%, and thresholds tuned by validation F1 for want of, in the 2017 phrasing, an "informed criterion" (§4). Only in 2026 does the cost term begin to reappear, in fragments (false-alarm pricing, measured cognitive load, social violation, compute budgets) that no work unifies into one decision, and none of which conditions on what the user is doing right now.

We survey this terrain through a five-layer taxonomy (Figure 1): **signals** (what an always-on agent can watch, §3), **decision** (when to act, §4), **action** (what it does, deliberately brief because it is the mature layer, §5), **memory** (what it knows about this user, §6), and **evaluation** (whether the intervention was good, and whether the system improves, §7). The layers are not symmetric in health. Signals are rich and getting richer, yet are spent on intent understanding rather than timing; the one signal always-on assistance cannot do without, elapsed time itself, is one current models demonstrably fail to use even when handed timestamps. The action layer is commodity. The decision layer is the bottleneck, and the evaluation layer is what gates it: reinforcement learning needs a reward, a reward for intervention quality presupposes a measurement of intervention quality, and no such measurement exists for the open world. We compress this into a slogan the rest of the paper defends: *reward* ⊆ *eval*.

Adjacent literatures mark our boundaries. Deng et al. [1] cover proactivity in conversational systems: what to say, and toward what goal, once speaking is licensed. We cover whether to speak at all. Li et al. [2] map egocentric procedural assistance, the task landscape our target systems live in; we cut across it on the single axis of timing and its cost. Mobile health owns a mature timing formalism: the JITAI framework [3] decomposes the decision into decision points, tailoring variables, intervention options with an explicit *provide nothing* action, and threshold decision rules, and names intervention burden and fatigue as first-class costs. But its thresholds are theory-set free parameters, neither expectation is estimated, and its setting is a health prompt to a patient rather than an open-ended assistant acting mid-task. The closest prior survey is Pejović and Musolesi

[4] on anticipatory mobile computing, which devotes one subsection to opportune moments for interruption inside a broader treatment of mobile sensing and context prediction; it neither organizes that literature around a decision rule nor prices the intervention, and it predates always-on LLM assistants entirely. On our axis (intervention timing for always-on, LLM-era assistants, organized by the cost of speaking), this is, to our knowledge, the first dedicated survey.

Our contributions: (1) a five-layer taxonomy of intervention timing, maintained as a living public map;[1] (2) a reconnection of the 1999–2017 interruptibility formalisms to the 2024–2026 proactive-agent wave, with the port made explicit formalism-by-formalism (Table 2); (3) an assessment of the 2026 state: the cost term is modeled in fragments with no unified θ, three mutually uncited papers converge on a cheap-gate/expensive-actor architecture, and the deployed state of the art avoids the decision entirely; (4) the design of a benchmark for open-world intervention timing with an explicit cost term (§8); and (5) six open problems, ordered by dependency (§9). Throughout, every quantitative claim is traceable to a numbered table or section of its source; where a paper's prose and its tables disagree, we cite the table, and where a paper's equations and its released code disagree, we say so.

Method and scope. The corpus was assembled in two passes rather than by a single database query, reflecting the two lineages. The 1999–2017 interruptibility literature was seeded from its canonical works and closed under backward and forward citation chasing through the ACM Digital Library and Google Scholar. The 2024–2026 wave was collected by keyword search over arXiv (cs.HC, cs.AI, cs.CV) and the ACM and IEEE libraries, using *proactive agent*, *intervention timing*, *interruptibility*, *when to speak*, and their variants as query terms. It too was closed under citation chasing, and is maintained continuously as the living map of contribution (1). Inclusion required that a work make, measure, or price the unprompted speak-or-stay-silent decision of an always-on assistant, or supply a formalism or measurement portable to that decision; conversational proactivity inside an already-licensed turn and notification-content optimization fall outside the boundary drawn above. One consequence of surveying a two-year-old wave should be stated plainly: of the 47 works cited, 23 are arXiv-only preprints from 2024–2026 that have not passed peer review, including load-bearing ones (PRISM, WakeAnchor, Pro^{2}Bench). This is why the survey's citation discipline is what it is: numbers are cited to tables, not prose, and checked against released code where code exists.

2 The Lineage: Two Decades of Knowing When Not to Interrupt

Before any LLM could act on a user's behalf, a research community spent nearly twenty years on the other half of the proactive-agent problem: what does it cost to interrupt a person, and how should that cost be weighed? This literature is rarely cited by current proactive-agent papers, which is a loss twice over. It contains the only rigorous formalizations of the interruption cost term to date, and its trajectory, from full decision theory (1999) to learned cost estimation (2003–2005) to cost-free engagement prediction (2017), explains precisely how the modern field arrived at capable actors with no cost model. We reconstruct that trajectory here, because the formalisms port directly; Figure 2 lays out both lineages on one axis, and Table 1 summarizes what each generation had and what it lacked.

[1] https://github.com/tao-hpu/awesome-proactive-agents

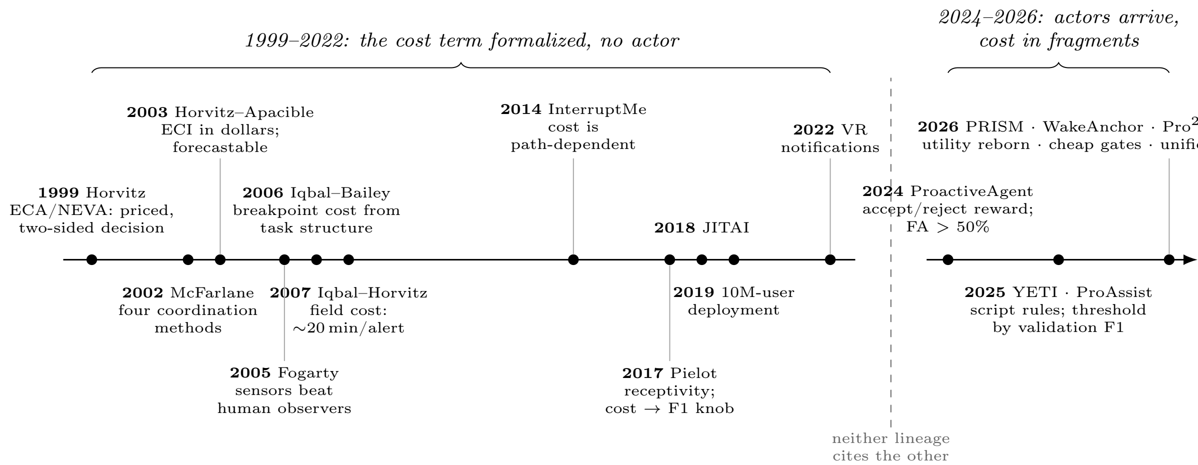

Figure 2: Two lineages on one axis. The interruptibility literature (left) formalized the cost of speaking (expected cost over an inferred attentional state, a learned price in dollars, sensor estimators that beat human observers) and had no actor capable of delivering benefit. The proactive-agent wave (right) has the actor and rebuilt the decision layer from scratch; the cost term reappears only in 2026, in fragments. No work on the right cites the formalisms on the left. Note the right segment's expanded time scale.

2.1 The expected-utility era: interruption as a priced decision

Horvitz et al. [5] remains the most complete statement of intervention timing as a decision problem. Its starting premise is the one this survey argues the field must recover: "human attention is the most valuable and scarcest commodity in human-computer interaction" (§1). The framework prices both sides of the decision.

On the cost side, the system maintains a Bayesian network over the user's *attentional state* (inferred from calendar, time of day, ambient audio, the application in focus, and keyboard/mouse activity streams) and computes the expected cost of alerting as an expectation over that latent state:

$$\mathrm{ECA} \;=\; \sum_j C_a(A_i, F_j)\, p(F_j \mid E_a), \tag{2}$$

where $C_a(A_i, F_j)$ is the cost of alerting via modality A_i while the user is in attentional state F_j. Cost is not a constant, not a heuristic, and not a threshold: it is a function of what the user is doing right now, marginalized over the system's uncertainty about it.

On the benefit side, the framework is equally careful about the value of *silence*. The expected cost of delayed action discounts the value of information with time and, critically, is computed against a model of when the user would encounter the information *anyway* (their unprompted inbox-inspection interval). The alert decision is then a net-value rule:

$$\text{alert} \iff \mathrm{NEVA} = \mathrm{EVTA} - \mathrm{ECA} > 0, \tag{3}$$

where EVTA is the expected value of transmitting the alert now versus letting the user find the content on their own. Nor was the benefit estimate feeding EVTA a toy: the system's inferred message criticality correlated 0.9 with expert hand scores (their §5). Two properties of this rule deserve emphasis because no current proactive agent has either. First, *waiting is a first-class action* with a computable value, not a fallback. Second, the framework prices *bundling*: alerting about n

messages at once amortizes one distraction across several items (their Eq. 8), anticipating by 25 years the question of whether an always-on assistant should batch its interventions.

The limitation is equally instructive. The action space is notification mediation only: alert, defer, chunk, choose a modality, forward to a pager. The system decides *whether to speak, never what to do*, so every unit of benefit in EVTA is ultimately delivered by the user's own subsequent attention. The decision theory was complete; the actor was missing.

The same year's CHI sister paper [6] supplies the decision geometry that Eq. 3 leaves implicit. Utilities over the four outcomes of {act, don't act} $\times$ {goal, no goal} yield expected-utility lines that cross at a threshold probability p^*, and the paper's central observation is that the threshold is not a constant: "the utility of unwanted action can diminish significantly with increases in the depth of a user's focus on another task. Such a reduction in the value of action leads to a higher probability threshold" (§Expected Utility and Thresholds). Adding dialog as a third action with its own utility line produces *two* thresholds and a three-way policy (stay silent, ask, act), placing the cheap clarifying question in a principled middle band a quarter-century before modern assistants rediscovered it as a UX pattern (§3.3). The paper evaluates nothing; it is principles plus a prototype. But it is where θ's license to move first appears, and its opening diagnosis of agent failure ("poor guessing about the goals and needs of users, inadequate consideration of the costs and benefits of automated action, poor timing of action") reads unchanged as a review of the 2026 systems in §4.

Four years later, Horvitz and Apacible [7] removed the framework's main scaling obstacle (hand-built attention models) by learning the cost term from data. Subjects tagged video of their own working hours with interruptibility levels and assigned each level a dollar value ("willingness to pay to avoid a disruption," borrowed from medical decision analysis), and a Bayesian network learned to predict the expected cost of interruption from desktop events, calendar, audio, and head-pose sensing. The result is a cost estimate in a real currency, computable at run time (accuracy .73/.64 for two subjects, their Table 1), and, remarkably, *forecastable*: the model predicts the time until the user will next be cheaply interruptible and remain so for at least five minutes. Forecasting turns "wait" from an indefinite deferral into a plannable action with a horizon. The same paper also contains the era's sharpest warning about personalization: models transferred across users scored .28/.32, *below* majority-class guessing (their Table 3). A timing policy is not a global object.

2.2 The coordination question: who decides the moment

Before the cost estimator could be built, the field settled a prior question: interruption timing is a *coordination* choice, and a system that decides the moment is only one of four ways to make it. McFarlane [8] taxonomized them and compared all four in a 36-participant dual-task experiment: *immediate* (the system interrupts now), *negotiated* (the system announces; the user picks the moment), *mediated* (a broker holds the interruption until a cheap moment), and *scheduled* (a fixed cycle). Every expected-utility gate in this survey, from Eq. 3 to PRISM, occupies the *mediated* cell. The result is a standing caution for that cell: negotiated won on most of the twelve design goals, immediate won only on timeliness of the interrupting task, and the paper's own summary is that "there is no one 'best' method for coordinating interruptions for all kinds of human performance. There are instead, tradeoffs" (§6). Two specifics outlived the study. First, cost is multi-dimensional: accuracy, efficiency, completeness, and timeliness rank the four methods *differently* (their Fig. 16). Any scalar θ therefore silently encodes a weighting over incommensurable costs. Second, its assertion that interruptions delivered "at the lull points of the continuous task" let users perform better on both tasks (§5.4) is the empirical license for the breakpoint line below. The mediator McFarlane tested was a crude visible-workload count, so its middling showing bounds naive mediators, not the strategy; but the negotiated baseline (announce and let the user choose) remains one the modern proactive-agent literature almost never runs.

2.3 The sensor era: the cost estimator isolated

Fogarty et al. [9] supplies the empirical bounds for the cost-side estimator. From 602 hours of recorded office life and 672 experience-sampled self-reports, simulated sensors predict the binary "highly non-interruptible" state at up to 82.4% accuracy, significantly *better than human observers* shown the same footage (76.9%, itself only modestly above the 70.6% base rate). Two findings matter for any always-on agent. First, the useful signals are shallow: all 30 top features by information gain relate to talking or the telephone, and a single "is anyone talking" sensor alone reaches 75.9%. Second, and central to this survey's thesis, the paper deliberately contains *no decision rule*. The trade against benefit is named and deferred, four separate times: "Because systems do not have a mechanism for weighing the importance of information against the appropriateness of an interruption, people are forced into extremes of either allowing all interruptions or forbidding all interruptions" (§1). Here the cost term stands fully isolated from any decision. It is the exact seam along which, two decades later, the LLM era would pick up the actor and drop the cost.

2.4 Breakpoints: cost from task structure, and its measured magnitude

Iqbal and Bailey [10] relocated the cost term: not in the ambient signals around the user, but in *where the user is inside their own goal hierarchy.* Three structural features of a task decomposition (the depth of the boundary being crossed, how much data must be carried over it, and the difficulty of the *next* subtask) predict a 3.7× spread in resumption lag across subtask boundaries (464 ms at the cheapest class of boundary, 1,012 at the middle, 1,702 at the dearest, their §Differences in Resumption Lag), with no sensor at all: the model is legible offline, from the task structure alone. The three surviving predictors are all workload proxies; "COI depends more on the characteristics that reflect current and prospective allocation of mental resources (workload) than on those that reflect temporal position or cue availability" (§Determine the Most Predictive Factors). Two boundaries deserve marking. The model contains no benefit term by design: its output is "sent to a broader reasoning framework" that never arrived. Its authors also note the same three-class ceiling Fogarty found, suggesting a hard bound on how fine-grained any sensorless cost estimate can honestly claim to be. The pointed irony for §4: the script-following agents of 2025–2026 possess exactly the task decomposition this model needs, and use it only to detect *deviation*, never to price the moment.

Iqbal and Horvitz [11] then measured what an interruption actually costs in the field, and the magnitude indicts every casual cost assumption since. Across 27 information workers, two weeks, and 2,267 logged hours, alerts arrived 3.74 per hour, and participants "spent on average nearly 10 minutes on switches caused by alerts, and spent on average another 10 to 15 minutes (depending on the type of interruption) before returning to focused activity on the disrupted task" (§Summary). That is a twenty-minute arc for responses that took seconds, with a heavy tail: 27% of alerts left the suspended window unresumed after two hours, and tasks engaged less than five minutes before the alert had a 10% chance of not being resumed within two hours at all. The study also caught users doing breakpoint deferral *manually*: in the seconds between an alert's arrival and the switch, paragraph-completion rates spiked from 0.78 to 10.9–12.8 per minute, as users raced to reach a subtask boundary that a mediated system could have waited for on their behalf. Any model that prices an interruption at the duration of its response is off by two orders of magnitude, and any gate that optimizes expected cost under a symmetric loss will systematically under-price the tail.

2.5 The mobile era: a sharper target, a blunter theory

Pejović and Musolesi [12] took the gate onto the phone and into the wild, and surfaced the property every instantaneous-context gate since has ignored: interruption cost is *path-dependent*. In a month-long deployment, sentiment toward an isolated notification was more than twice as favorable as toward one preceded by eight others in the prior two hours; and because favorable context persists, a context-driven gate fires back-to-back and "exhaust[s] the interruptibility of the user", which is actively worse than context-oblivious random scheduling on that axis (§Amount of Interruptibility). The same paper states the cost asymmetry as a metric choice a decade before anyone derived it: "We are predominantly interested in the precision as our main goal is to design a 'calm' system that does not interrupt at inappropriate times, and we assume that opportunities for interruption are ample" (§Evaluation of Interruptibility Models). That is $C_{\mathrm{FA}} \gg C_{\mathrm{FN}}$, assumed rather than estimated, enforced through the choice of what to score. The gate itself is a per-user online classifier with no utility model anywhere, and its authors' own future-work sentence (models that take interruption load into account) names a gap the following decade left open.

Pielot et al. [13] marks both the lineage's methodological peak and its theoretical low point. The target improves on everything before it. Interruptibility (will a notification *attract* attention?) is the wrong question for proactivity; the right one is *receptivity*: "predicting if users are interruptible ... does not imply that their attention will be held as well, i.e., result into engagement" (§1). In a four-week field study (337 users, over 120 million phone events, 78,930 notifications), an XGBoost model over 197 ambient phone features predicts whether the user will voluntarily consume optional content at this moment. The headline results: a five-fold precision lift over base rate once *per-user history* is added, while using only 10.6% of possible notification moments. Four features summarizing the user's own past responses outperform all 197 context features combined. Receptivity is learnable from ambient signals, and it is overwhelmingly personal.

But observe what has happened to the cost term. There is no utility model and no cost function, only a classifier and a threshold. Where 1999 computed ECA against EVTA, 2017 selects the F1 score, explicitly for want of anything better: "As we do not have an informed criterion on the relative importance of precision and recall ... we used the F1 score as primary evaluation metric" (§5.4). Interruption cost has degraded into a precision/recall knob plus a count of pings; and the optimization target has partly drifted from the user's welfare to the product's conversion rate. The strongest model also achieves its precision by effectively writing off ∼15% of users as never-opportune, a distributional failure mode no aggregate metric surfaces. The lineage thus ends with the field able to *find* good moments but no longer *pricing* bad ones.

2.6 The plateau, and what it explains

After 2017 the interruptibility line did not so much die as plateau and disperse. It produced its own capstone survey of attention management systems [14], whose vocabulary is the one in which the cost term lives: attention fragmentation, breakpoints, deferral. That vocabulary also crossed into mobile health, where the JITAI framework [3] gives intervention timing its most complete component decomposition (decision points, tailoring variables, an explicit *provide nothing* option, burden and fatigue as named costs) while leaving the thresholds theory-set and both expectations unestimated. It shipped. Breakpoint-based notification deferral, which is Iqbal and Bailey [10] operationalized at population scale, was evaluated in a 382,518-user, 28-day deployment study reporting up to a 60.7% click-rate increase, and was then enabled for all of Yahoo! JAPAN's more than ten million Android users [15]; the 2018 journal predecessor had already run a >680,000-user study reporting a 49.7% reduction in response time [16]. This is the lineage's population-scale proof, and the direct answer to the "offline replay, never deployed" limitation of Pielot's study. And it crossed into the headset: opportune-moment prediction for notifications inside VR [17], four years before the current proactive-MR wave, which cites none of it.

What the line never did was meet an actor. The benefit ceiling explains the plateau: every system in this lineage could only mediate the delivery of information a human would still have to act on. When the maximum upside of a perfectly timed intervention is "the user reads a message slightly earlier," investment in the cost model has nowhere to compound. The deep formalisms were built for an actor that did not exist.

The actor now exists. GUI agents complete multi-step tasks against ground-truth verifiable goals (§5); egocentric perception resolves user intent at high accuracy (§3). But the works that inherited the *proactive* ambition (§4) rebuilt the decision layer from scratch, without the inheritance: rule-based triggers with no cost term, or learned relevance models with no state-dependent interruption price. Only in 2026 does the cost term begin to reappear, and then only in fragments; PRISM's asymmetric false-alarm pricing is the closest single rebuild (§4.4). None of these works cites the formalisms above. The full synthesis this history points to remains unattempted: Horvitz's expectation-over-attention cost, Fogarty's shallow-sensor estimators, and Pielot's personal receptivity histories, attached to a modern actor whose benefit term is no longer bounded by the user's own attention. Table 2 makes the port explicit, formalism by formalism; Sections 4 and 8 take up what executing it requires.

3 Signals: Rich Perception, Aimed at the Wrong Question

In three years the input available to an always-on assistant has gone from a keystroke log and an ambient microphone to synchronized streams of egocentric video, 30 Hz gaze scanpaths, hand and finger kinematics, IMU, GPS, and duplex audio, all on shipping consumer glasses. Almost none of that richness is spent on timing. Signals are consumed to answer *what is the user doing* and *what do they mean*; the question *is this the moment to speak* is answered, where at all, by a hand-set constant. The pre-LLM sensor floor already showed how little perception a timing estimator needs. In Fogarty et al. [9], a single "is anyone talking" sensor alone predicted the "highly non-interruptible" state at 75.9%, statistically indistinguishable from human observers (§2.3). That makes the pattern worse, not better: the field acquired far richer signals and did not point them at the decision.

3.1 Gaze and eye–hand: the richest signal, fired on a metronome

Wang et al. [24] assembles the most detailed embodied signal set in this literature: 30 Hz gaze reduced to fixation-and-shift states over named objects, hand position and rotation in a body-relative frame, and five finger shapes, all translated into language an LLM can reason over. Its ablation is the most useful signal-level result available to a timing model: gaze alone recovers the correct intent 30.2% of the time at rank 1, while gaze *plus* hand motion reaches 58.3%, with top-3 rising 63.5% → 75.0% (their Table III). Gaze is not sufficient; gaze conditioned on manual action nearly doubles it.

What the system does with that signal is the emblem of this section. Sampling is a constant: "Based on pilot exploration indicating that users can express their intent within 3 seconds, we set a fixed data acquisition window of 3 seconds" (§IV-A). No continuous inference, no early exit, no confidence threshold deciding whether *now* is the moment. And the agent never acts on its own reading: "Users select from LLM's ranked predictions, providing explicit feedback for intent determination" (§IV-B). The false-positive problem is not solved but sidestepped. The human is the gate. The best embodied signals in the field, fired on a metronome and routed through a confirmation dialog, with no interruption cost, no false-alarm penalty, and no timing metric in the paper.

Work	Cost formalization	Action space	Key number	Missing half
Horvitz et al. [5]	ECA over inferred attentional state; alert iff NEVA $>$ 0; time-discounted silence; bundling	notification mediation only	criticality vs. hand-score corr. 0.9	actor
Horvitz and Apacible [7]	learned ECI in dollars; interruptibility *forecasting*	notification modality/timing	COI accuracy .73/.64; transfer .28/.32	actor; benefit side "out of scope"
McFarlane [8]	none (cost measured as an outcome across nine metrics)	four coordination policies compared	negotiated best overall; immediate best only on timeliness	a mediator worth using
Fogarty et al. [9]	none (deliberately deferred)	none (estimate only)	82.4% $>$ human 76.9%; one talk-sensor 75.9%	the decision itself
Iqbal and Bailey [10]	COI classes from task structure alone; no benefit term by design	none (estimate handed upward)	3.7$\times$ lag spread 464$\rightarrow$1,702 ms; 3-class ceiling	the decision; any sensor
Iqbal and Horvitz [11]	none (pure field measurement)	none	$\sim$20–25 min true cost per alert; 27% $>$2 h tail	any model at all
Pielot et al. [13]	none (F1 knob + ping count)	when to fire a push	5$\times$ precision lift from personal history; 10.6% of pings	cost model; actor

Table 1: What each generation of the interruptibility lineage had, and what it lacked. Bridge works 2018–2022 [14, 15, 17] refine rather than alter the pattern: ever-better cost estimation, still no actor.

3.2 Scene change: the one signal actually wired to timing

Bandyopadhyay et al. [25] is the counterexample, and it is instructive that it works with the poorest signals. Two scalars over egocentric RGB gate the intervention directly: frame-to-frame structural similarity (SSIM) and the change in per-second object count. The system fires at frames passing an SSIM filter ($\tau = 0.9$) that sit at a local extremum of the object-count delta. Those features cost 21 MB and 20 MB against 137,408 MB for depth and 617 MB for gaze (their Table 1), and Global YETI still reaches 41.86 precision / 88.31 recall / 56.17 F on HoloAssist intervention detection, against 13.93/33.33/19.65 for the RGB baseline.

But 41.86 precision over-interrupts roughly seven times for every five correct calls, because cost is rate-limited, never modeled: a 1 s minimum gap "ensur[es] that the AI agent does not intervene too frequently" (§3.6). That is a knob, not a price. Nor is the target defined: "As there is no empiric measurement of when the 'best' precise moment to intervene is, we use a window of five seconds around HoloAssist labelled ground-truth proactive intervention starting time-stamps" (§4.2). The one system that routes a signal into a timing decision has to invent its own ground truth to score it.

1999–2017 formalism	What it supplies	Where the 2026 wave lacks it	The port
ECA: expected cost of alerting over an *inferred* attentional state (Eq. 2) [5]	a state-dependent interruption price	boolean vetoes and rate limits: EgoSocial's SFD "global veto," YETI's 1 s minimum gap (§3)	replace the boolean with $\mathbb{E}[\text{cost}]$ marginalized over the busy-state posterior the perception stack already computes
NEVA: two-sided pricing, silence time-discounted as a first-class action (Eq. 3) [5]	a reward that cannot be gamed by mutism	one-sided penalty feedback collapses recall $98.11 \rightarrow 56.76$ [18] (§4)	add an explicit cost-of-silence term before any preference optimization
ECI: per-user interruption cost learned in dollars; transfer scores .28/.32, below majority-class guessing [7]	a measurable, personal C	PRISM's $C_{\text{FA}}, C_{\text{FN}}$ are population-level free knobs, swept not measured [19]	estimate C per user from interaction traces; never transfer it across users
Shallow-sensor interruptibility estimation: one talk-sensor reaches 75.9% [9]	a cost-side gate cheap enough to run always-on	cheap gates price *compute*, not interruptibility [20, 21]	point the cheap gate at the user's state, not only at relevance and budget
Personal receptivity history: 4 self-history features beat all 197 context features [13]	the initialization for a per-user θ	per-user dynamic thresholds exist only in a 20-item lab probe [22]	make the per-user scalar the *only* learned component, updated from L0/L1 feedback (§7)
Breakpoint COI from task structure: three plan features predict a 3.7× cost spread, offline [10]	an interruption price legible from the agent's own plan, at zero sensing cost	script-following agents use their task decomposition to detect deviation, never to price the moment (§4)	annotate the plan's subtask boundaries with COI classes before execution
Delayed-action counterfactual: value of alerting now vs. when the user would find out anyway [5]	per-intervention credit without waiting for outcomes	no benchmark attributes a downstream outcome to a specific intervention (§9)	estimate the "without-me" baseline from the behavioral traces Tang et al. [23] already collects

Table 2: The porting guide: what each recovered formalism supplies, the 2026 gap it addresses, and the concrete port. Each row is expanded in the section cited in its third column.

3.3 Always-on streams: perception solved, initiation untouched

Continuous capture is now engineering rather than research, and three 2026 systems leave initiation empty regardless. Liu et al. [26] streams Meta Ray-Ban video at roughly 1 fps plus continuous audio to a live multimodal model with no wake word, logging 555 voice-initiated interactions across 55 participant-days in the wild. Yet "VisionClaw is limited to reactive interaction, meaning the agent acts only when the user initiates a task through voice input" (§8). A continuous stream and a capable actuator, every action still routed through an utterance, because nothing decides when to speak first. Yang et al. [27] adds gaze by 3D ray-casting and the strongest long-horizon memory design here, then builds an uncertainty estimator and wires it only to a clarifying question, fired "when the LLM detects high semantic uncertainty or conflicting interpretations" (§3.4): disambiguation *inside* a turn the user already opened, not a decision to open one.

Yang et al. [21] comes closest to a timing architecture. Its tiered cascade runs across 92 hours of in-the-wild capture on commercial glasses (with inference offloaded over the network, and external power in the prototype) from 20 volunteers in five cities. The cascade layers always-on IMU, GPS, and audio; vision escalated on a 5 s/60 s dual-rate duty cycle when the user moves or nears a relevant point of interest; a 20 s suppression interval; and ORB focus ($\tau_f = 0.2$) and BERT redundancy ($\tau_t = 0.5$) gates before delivery. Feeding the model's own need prediction back into the sampling rate is load-bearing (21.2 F1 points, against 3.1 for the cheap-cue trigger). But every threshold is hand-tuned, and the paper prices *sensing*, not interruption. It also exposes how loose the field's timing ground truth is: assistance is annotated as "acceptable within a temporal window rather than at a single precise timestamp" (§5.2.1), and "the average participant-annotated acceptable window for proactive assistance is only 2.8 minutes" (§5.5). *Only* 2.8 minutes. At that tolerance, over 85% of participants agreeing the timing was appropriate is close to unfalsifiable, and what is measured is coarse opportunity detection, not timing.

3.4 Gaze as anticipation, and the warning attached to it

Lee et al. [28] is the one work treating gaze as a timing signal rather than an intent signal: "eye gaze is the most direct and reliable indicator of where a user is attending, what they are processing, and what they are likely to do next" (§1). It formalizes the distinction a proactive agent exists to exploit: O_{fov}, what the user has fixated, versus O_{out}, what is visible in the same frame and *not* fixated. It then turns that distinction into an Object Appearance Alert task that fires "when the specified object first appears in the peripheral region" (§4.2.3): the canonical should-I-speak trigger, noticing what the user has not noticed.

Models cannot do it. Against a human ceiling of 0.827 overall, the best model (GPT-4o) reaches 0.535, and the collapse concentrates on anticipation: GPT-4o's proactive-family average is 0.373 against a human 0.7725, with Object Appearance Alert at 0.149; InternVL3.5-8B scores 0.051, essentially never noticing a newly appearing object. And handing the model the gaze signal does not help. On Qwen2.5-VL-7B the gaze-free baseline averages 0.446 while textual and visual gaze prompts both give 0.429 and a salience map 0.454, so "none of the prompting strategies consistently outperform the gaze-free baseline" (§5.3); the purpose-built gaze-native model scores 0.223. The authors' diagnosis is the right one: the bottleneck is "the absence of explicit mechanisms for linking gaze shifts to predictive temporal reasoning" (§5.2).

3.5 The temporal signal: the slot is occupied, and it is failing

The sharpest evidence comes from the work that strips vision away entirely. Cheng et al. [29] isolates timing from perception: the agent observed X at t_0, it is now $t_0 + \Delta$, and a tool call would refresh it. Is the old observation still good enough to act on? Ground truth is a human preference vote (3,016 samples, Krippendorff's $\alpha = 0.8574$), scored by a normalized alignment rate where 50% is chance. Frontier agents sit barely above the coin flip: "no model achieving a normalized alignment rate better than 65% when given time stamp information" (abstract); without timestamps the ceiling is just above 55%.

Three diagnostics turn a low score into a structural claim. Timestamps make models *twitchier*, not better calibrated: "one would expect human-like temporal awareness to manifest as an increased attempt rate on prefer-Tool samples and a decreased attempt rate on prefer-noTool samples. However, we observe that attempt rates increase across both subsets for most models" (§4.1). The clock never reaches the reasoning: across Qwen3 models a timestamp appears in 1.03%–3.18% of traces and the word "timestamp" in 0.17%–1.43%, and absent a real timing signal the agents confabulate one, using turn count as a staleness proxy "regardless of whether timestamp information is provided" (§4.2). Decisively, the ablation that hands the model the pre-computed elapsed delta, so that no arithmetic is required: "for most models, even with explicit Δt, there is no alignment increase... the improvement is marginal (no more than 1%-4%)" (App. B). The bottleneck is not computing how long it has been; it is knowing what a duration *means* for this environment, which takes a volatility model nothing is trained to build.

Beside StreamGaze's net-zero gaze result, the pattern across two unrelated modalities is one claim: **supplying the timing signal does not produce timing behavior, because nothing converts signal into decision.** Which is why TicToc's constructive half matters as much as its negative one. DPO on human preferences over *when refreshing was worth it* (trained on 50 scenarios, evaluated on 26 held out) lifts alignment from roughly 61% to 89% (Qwen3-8B), 50% to 89% (Llama-3.2-3B), and 55% to 85% (Ministral-8B), while prompting barely moves. Timing is learnable, and learnable specifically from preference data *about the timing decision itself*. That is the supervision no egocentric proactive benchmark in §7 collects, and the constraint §8 takes as its starting point.

4 Decision: When to Act

This is the survey's core layer, and the one where the field's inheritance problem is most visible. Current decision mechanisms fall into three groups: rules without a cost term, learned relevance without a cost term, and systems that defer the decision to the user. A fourth, decision-theoretic group reappears only in 2026. None of the works below cites the formalisms of §2.

4.1 Rule-based: deviation implies intervention

YETI [25] is the cleanest modern statement of rule-based timing. Two signal streams decide when to speak during procedural AR tasks: structural frame differencing (SSIM) detects scene change, and an alignment check compares the user's current action against the expected task step. Deviation from the script is treated as sufficient evidence to intervene. There is no cost term: the rule fires when the mismatch fires, regardless of what interrupting costs at that moment. The design is honest about its dependency. It works because HoloAssist-style procedural tasks come with a script that defines "off track." Pro^{2}Assist [30] is the 2026 successor: continuous step tracking from camera and IMU across long-horizon procedures, with timing accuracy reported as a headline metric (up to 2.29× baselines). The rule sharpens, the scenario assumption stays: no script, no deviation, no trigger. Open-world daily life offers no script (§9).

4.2 Learned relevance: a classifier and a threshold

The learned group replaces the rule with a model of whether help would be *relevant*, but keeps the same missing term. Lu et al. [18] train a reward model on human accept/reject labels over 6,790 generated events (ProactiveBench); the reward model itself is strong (F1 91.80, and 100% agreement with humans on false-detection cases where GPT-4o scores 0%), and fine-tuned agents reach F1 66.47. But the agent-side numbers expose the cost blindness: the strongest proprietary models sit at false-alarm rates above 50% (GPT-4o 51.85, GPT-4o-mini 64.73, their Table 3). These are capable actors volunteering help at moments users reject more often than not.

The paper's own ablation then delivers, inadvertently, the survey's cleanest demonstration of why a one-sided penalty is not a cost model. When reward-model feedback is added at inference time to punish bad proposals, false alarms drop, and recall collapses (GPT-4o: 98.11 to 56.76): "models stay silent once they receive feedback" (their §4.3). This is exactly the failure the 1999 formalism is built to prevent: Eq. 3 prices *both* the interruption and the delayed help, so silence is a costed action rather than a safe default. Penalize false alarms without valuing timely help, and the optimum is mutism. The field re-discovered, empirically, why the decision needs two terms.

ProAssist [31] moves the learned decision to streaming egocentric video: at every frame the model either stays silent or begins an utterance, with a speaking threshold θ on the silence probability. The authors report that "model performance is highly sensitive to θ, with a clear precision-recall tradeoff" (their §6.1), and θ is chosen by validation F1. This is the same move as Pielot et al. [13], seven years later and one layer deeper in the stack: the cost of interruption compressed into a single scalar hyperparameter, selected for want of an "informed criterion." What the system calls conservative versus impulsive talking styles is a utility trade-off with the utilities left implicit.

4.3 The deployed reality: always-on perception, reactive action

The systems closest to production sidestep the decision entirely. VisionClaw [26], an agent on commercial smart glasses, is always-on in perception but strictly reactive in action: "the agent acts only when the user initiates a task through voice input" (their §8); the continuous visual stream grounds the user's request but never initiates one. Proactivity is named as future work, with the authors noting it "remains unclear how proactivity can be integrated with general-purpose agents." SIAgent's fixed three-second window with mandatory human confirmation (§3) is the same pattern in VR. The most deployable systems of 2026 solve the timing problem by not having one: the human remains the gate. This is a rational engineering response to an unpriced risk, and a measure of how far the decision layer lags the action layer.

4.4 Utility reborn: PRISM

PRISM [19] is, to our knowledge, the first LLM-era system to rebuild the intervention decision as explicit expected-cost minimization. The gate is

$$\text{intervene} \iff p_{\text{accept}} \;\geq\; \tau(p_{\text{need}}) \;\triangleq\; \frac{C_{\text{FA}}}{C_{\text{FA}} + p_{\text{need}} \cdot C_{\text{FN}}}, \tag{4}$$

derived by Bayes-risk minimization over the asymmetric costs of a false alarm (C_{FA}) and a missed intervention (C_{FN}); the comparative statics are the right ones ("higher burden C_{FA} tightens the gate, while higher risk C_{FN} or higher need p_{need} relax it," their App. A), and the probabilities are post-hoc temperature-calibrated rather than taken raw from the model. On ProactiveBench this cuts the strongest prior agent's false-alarm rate from 50.22% to 22.94% while lifting F1 from 66.47 to 86.61 (their Table 1). Structurally, Eq. 4 is Eq. 3 reborn with learned probabilities in place of elicited ones.

Three qualifications keep this a beginning rather than a synthesis. First, the cost pair $(C_{\mathrm{FA}}, C_{\mathrm{FN}})$ is fixed per deployment, not a function of the user's current attentional state; the central insight of 1999 (Eq. 2) is still absent. Second, the setting is desktop event streams, not open-world MR. Third, a reproducibility caveat our review surfaced: the released code's default threshold implementation is a reciprocal-shaped variant of the paper's Eq. 3.1 ($\tau = C_{\mathrm{FN}} q_e / (C_{\mathrm{FA}} + C_{\mathrm{FN}} q_e)$, with all reported results using this form), and the paper does not reconcile the two. Cite the equation, but check the code.

4.5 A convergent architecture: cheap gate, expensive actor

Three 2026 papers, mutually uncited, converge on the same architecture: a lightweight always-on gate decides *whether* this is a moment, and the expensive model is invoked only afterward. PRISM gates its slow, deliberative reasoner on borderline cases only (fire when $|p_{\mathrm{accept}} - \tau|$ falls inside a band). Liu et al. [20] push the argument to its limit: the trigger need not be an LLM at all. A temporal-graph encoder over structured $(actor, verb, object, timestamp)$ event streams emits a per-event trigger probability at 11–14 ms per event and a ∼220 MiB footprint, roughly 4–7× faster than every LLM-as-trigger configuration they test on servers and 12–83× on laptops. The LLM is called only after the trigger fires, to phrase the message. A pre-reasoning perception framework for mobile agents [32] makes the same move with a multimodal perceptor gating a reasoner, reporting lower false-trigger rates. The convergence matters for two reasons. Practically, always-on intervention on battery-powered devices makes the gate's compute cost part of the cost term. Conceptually, it separates the timing decision from the actor. That is precisely the decomposition the 1999–2005 literature assumed, rediscovered under an efficiency constraint rather than a decision-theoretic one.

4.6 Where the layer stands

Table 3 aligns the decision mechanisms. Reading it against §2: the rules of §4.1 have an actor and a trigger but no price; the learned models of §4.2 price relevance but not interruption, and collapse into silence the moment a one-sided penalty is applied; the deployed systems of §4.3 avoid the decision; PRISM restores the two-sided decision but with state-independent costs, on the desktop. The synthesis the lineage points to exists in no system we found: state-dependent interruption cost (Eq. 2), learned receptivity with personal history [13], and a calibrated two-sided gate (Eq. 4), in front of a modern actor, in the open world. §8 argues the missing prerequisite is not architecture but measurement.

5 Action: A Commodity Layer, and One Thing It Teaches Us

This section is deliberately the shortest in the survey, and its brevity is the argument. The capability to *do* the thing (operate a GUI, read a scene, ground an object in 3D) is no longer where the difficulty lies. Dedicated surveys already cover GUI agents and egocentric perception in depth; we do not duplicate them. We cover the action layer only far enough to establish one claim, that the bottleneck is not capability, and to extract one insight the timing literature has missed.

5.1 The bottleneck is not capability

Wang et al. [33] is the load-bearing citation, less for its scores than for the condition under which they were obtained. A multi-turn reinforcement-learning agent, trained through a data flywheel over a hybrid GUI/file-system/terminal environment, reaches 47.5 on OSWorld and 73.3 on AndroidWorld.

Work	**Decision rule**	**Cost term**	**Scenario**	**Key number**
Bandyopadhyay et al. [25]	script deviation $\Rightarrow$ intervene	none	scripted procedural AR	—
Xu et al. [30]	step-state rules over long horizons	none	scripted procedural AR	timing acc. up to 2.29× baselines
Lu et al. [18]	reward model on accept/reject	one-sided (penalty collapses recall $98 \to 57$)	desktop events	agent F1 66.47; false alarms > 50%
Zhang et al. [31]	per-frame speak/silent, threshold θ	compressed into θ by validation F1	streaming egocentric (offline replay)	dialogue F1 up to 36.25
Liu et al. [26]	none (user-initiated only)	n/a	in-the-wild glasses	median latency 12.2 s
Fu et al. [19]	calibrated $p_{\text{accept}} \geq \tau(p_{\text{need}})$, Bayes-risk derived	two-sided, fixed $(C_{\text{FA}}, C_{\text{FN}})$	desktop events	FA 50.22%→22.94%, F1 66.47→86.61
Liu et al. [20]	temporal-graph trigger, LLM only after firing	compute cost priced	event streams, on-device	11–14 ms/event; ∼220 MiB

Table 3: Decision mechanisms in LLM-era proactive systems. The two-sided, state-dependent, personalized decision of §2 appears nowhere in full.

Multi-turn RL over long-horizon agentic behavior demonstrably works, but only *when the reward has ground truth*. Task completion is checkable: the file was renamed or it was not. Every ingredient of a learned proactive policy is therefore in hand except the one that cannot be checked. Whether an intervention should have been made at 14:32 rather than 14:35, or not at all, has no oracle to verify it against, so the machinery that produced UI-TARS-2 cannot be pointed at timing. This is the argument §7 develops in full: reward $\subseteq$ eval, and evaluation is the gate.

The perceptual and geometric substrate is comparably mature. Guo et al. [34] pairs a 532M-parameter vision encoder with a 20B-active mixture-of-experts and is state of the art on 38 of 60 public benchmarks, GUI and agentic tasks included; Lin et al. [35] recovers any-view geometry from a single plain transformer with a depth-ray target, surpassing VGGT on pose and geometry and supplying the spatial grounding a physical-world intervention needs to point at the right object. None of this is contested, and none of it is what is missing. An agent that can see the workbench, localize the misplaced part, and execute the fix still has nothing to tell it whether to say so now.

5.2 Intervention is not binary

The one non-commodity insight in this layer comes from outside the egocentric literature entirely. Every proactive system surveyed here models a two-way choice (speak or stay silent) and then discovers it cannot price the cost of speaking. Lin et al. [36], from the spoken-dialogue community, supplies the missing middle rung.

Its benchmark defines a **backchannel** by rule rather than by model: a segment counts as backchanneling "if it meets the following criteria: (1) it has a short duration of less than 1 second; and (2) it contains fewer than two words" (§3.2). A **takeover** is anything else that is not silence. Critically, the backchannel is priced at exactly zero turn-control cost: "If the model merely responds with silence or a backchannel, no takeover is deemed to have occurred" (§3.2). Silence and backchannel are scored identically.

That definitional move converts the intervention decision from a threshold into a menu. The action space is graduated (*silence* $<$ *backchannel* $<$ *takeover*), and each rung carries a different price, which is exactly what a cost-aware policy needs: a way to act *now*, cheaply, when the expected value of a full intervention does not yet exceed its cost. The same paper shows how badly current models handle that menu. Backchannel timing sits at the ceiling of divergence: Jensen–Shannon divergence of 0.950 (dGSLM), 0.977 (Moshi), and 0.984 (Freeze-Omni) on a 0–1 scale where 1 is complete divergence. No model, in short, backchannels when humans do. Pause handling is worse: Moshi takes the floor on essentially every within-turn user pause (takeover rate 1.000 synthetic / 0.989 on Candor), pauses that "are often not intended to yield the turn but rather to maintain control" (§3.3.1). Only Freeze-Omni, which carries an explicit speaking/listening state-prediction module, is meaningfully better (0.287 on Candor). That is evidence worth carrying into §4: an explicit state component beats an implicit one.

Two caveats belong on the record. The backchannel's zero cost is *asserted by definition, not measured*: no experiment shows a backchannel is cheaper for a real user than a takeover, and the benchmark runs on offline replay throughout. And the takeover rate counts interruptions without pricing them: a model that cuts in during a trivial pause and one that cuts in mid-thought score identically. The graduated action space is the contribution; the cost of each rung remains, here as everywhere in this survey, unmeasured.

6 Memory: Timing Is Personal, and Nothing Remembers

The same intervention, at the same instant, is help for one user and noise for another. This is not a hypothesis; it is the most replicated empirical result in the lineage of §2. Horvitz and Apacible [7] found that interruptibility models transferred across users scored .28 and .32, *below* majority-class guessing (their Table 3). Pielot et al. [13] found the converse from the same fact: four features summarizing a user's *own* past responses outperformed all 197 ambient context features combined, yielding a five-fold precision lift over base rate. A timing policy is not a global object, and the personal history is the single most valuable feature any timing model can hold.

Meanwhile, a mature memory infrastructure has been built for LLM agents. And the two literatures do not cite each other. No proactive system surveyed here conditions its *timing* policy on long-term memory; no memory system exposes an interface a timing policy could query. This section documents the disjunction and names the interface that would close it.

6.1 The nearest intersection, and how it misses

Kim et al. [37] is the closest existing work to memory-conditioned intervention, and the precision of its miss is what makes it useful. It builds exactly the substrate a timing policy would need: a structured episodic store binding each past query and response to a space label, scene description, object-of-focus referent, temporal moment, and action intent, retrieved by reciprocal rank fusion. On top of it, a regulated-speech spectrum lets the user say progressively less: a full utterance, a micro-utterance ("sugar?"), or nothing at all. Memory reconstructs the missing intent. It works: intent-resolution accuracy is 95.4% for a full utterance, 86.7% for a partial one, and 83.3% for a *zero*-utterance button press, with a 49.8% reduction in words spoken.

It is also the only paper here that measures the *social* cost of speaking, on a 7-point difficulty scale in a seven-day field study: full utterances score 4.28 in public against 2.63 for partial and zero. That is a quantified interruption cost, levied entirely on the user's side. Memory infers *what the user meant*, never *when the system should speak*; the trigger remains a button press. The design goal is "Proactivity with User Controlled Disambiguation" (§2.2), where proactivity means predicting the content of an under-specified query. SpeechLess hands the field a personalized spatial memory substrate and a social-cost instrument, and points neither at the system's own utterances.

6.2 The one system with a personal threshold

Liu et al. [22] is, as far as we can find, the only proactive system that implements a per-user timing threshold and then ablates the timing itself. Physiological and behavioral signals (electrodermal tonic level; mouse direction flips, hover counts, hover time) feed a per-user regression whose output is compared to a threshold θ that moves after every item according to what happened last time: rising when declined help proved unnecessary, falling sharply (-4δ) when help was withheld and the user then answered wrong.

The ablation is the rare content-controlled timing manipulation. Content, model, sensing pipeline, and initial classifier are held identical; only the threshold-update rule varies across three conditions: aligned, sign-inverted, and random drift. Response accuracy: aligned 62%, random 55%, misaligned 51%, against a 41% calibration baseline (the abstract's headline "+21%" is measured against that baseline, not the controls; the honest condition effects are 7–10 points, LME -0.070 and -0.103). False negatives fall from 50.9% to 22.9%. The mechanism matters more than the magnitude: trigger *frequency* was statistically indistinguishable across conditions (14.03 vs. 11.50 vs. 11.19 helpers per person, $p = .328$), while *acceptance* rate diverged sharply (.676 vs. .388 vs. .444, $p < .001$).

Well-timed help is not more frequent help; it is help that gets taken up. The causal channel of timing runs through acceptance probability, precisely the quantity a cost-sensitive gate must calibrate (§4).

Two limits keep this from being a counterexample to the section thesis. The personalization is a scalar θ tuned by a six-branch hand-written rule, not a memory: nothing about *this user at this kind of moment* survives the session. And the asymmetry between missing a struggle and interrupting unnecessarily lives entirely in the hand-chosen multipliers ($\pm 4\delta$ versus $\pm\delta$), never derived or measured. The system also over-fires badly, firing on roughly 73% of trials at about 33% specificity, which a 20-item lab block tolerates and an all-day assistant would not.

6.3 What a timing policy would ask a memory system

The memory infrastructure exists. Packer et al. [38] introduced OS-style paged memory with explicit main-context/external-context transfer; Chhikara et al. [39] productionized a scalable long-term memory layer; Li et al. [40] treats memory as a first-class OS resource, unifying parametric, activation, and plaintext memory in a single addressable abstraction. All three are built around one query type: *retrieve the content relevant to the current request.* None exposes the query a timing policy actually needs.

That query is not about content. It is about the user's response history to *the agent's own past interventions*: at moments resembling this one (this place, this activity, this time of day), did this user accept, dismiss, or ignore what was offered, and what did the resulting interruption cost? Pielot's five-fold lift came from exactly four such features on a phone; Horvitz's transfer collapse says the alternative, a population-level policy, is worse than guessing. Interventions and their outcomes are the training data a timing policy needs, and the one class of event no current memory system is asked to store. Closing this gap requires nothing new in memory architecture: only that proactive systems write their own interventions and the user's reactions back into the store, and that the timing policy read them out. That the two literatures have not met is not a technical obstacle. It is an oversight, and the cheapest of the gaps in §8 to close.

7 Evaluation and Self-Improvement: The Gating Layer

Every other layer can be built today: signals are abundant (§3), the actor is a commodity (§5), and the decision rule has a two-decade-old formal statement waiting to be reused (§2). What cannot be built today is a *reward*. RL works for GUI agents because task completion is checkable; intervention timing has no verifier, and every attempt to manufacture one has produced a metric that scores a slice and leaves the rest free. Hence the containment this section argues: $\textit{reward} \subseteq \textit{eval}$. The objective a proactive agent can be trained against is bounded above by the evaluation the field can construct. Until intervention quality is measurable, RL for proactivity is not hard; it is undefined.

7.1 Benchmark dissection: eight measurements, eight blind spots

Wang et al. [41] supplies the perceptual substrate and then defines the decision away: "Intervention type prediction is to predict the intervention types given an input of a window of 1, 3, or 5 seconds before the intervention" (§4.2). The window is anchored on an intervention *known to occur*; the model classifies which of three types it will be, never whether to speak. On Table 8 (evaluated on 10% of the data) the best modality mix reaches overall precision/recall 48.75/47.75 against RGB's 47.92/46.09.[2] The nearest thing to a trigger, mistake detection, has precision 12.96 (RGB) and 12.5

[2]The prose claims an R+H+E row at 48.31/37.59 and a "+35 percentage point" gain over RGB; Table 8 contains no such row and its RGB precision is 47.92. We cite the table.

(hands) on the mistake class (Table 7): an agent gated on it would speak wrongly seven times in eight and violate no metric in the paper, because an instructor's utterance is free by construction.

Lu et al. [18] makes speak-or-stay-silent a first-class prediction. On ProactiveBench (136 scenarios; 6,790 training, 233 test events) the best system reaches F1 66.47 at a 50.22% false-alarm rate (half of all proposals unwanted) and contributes a learned reward model standing in for the cost judgment (F1 91.80; 100.00% agreement on false detections, against GPT-4o's 0.00%). The blind spot is temporal: no too-early, no too-late, no latency, no window, though §4.4 names "inappropriate proposing time" as a main failure mode. The objective seals it: "Our proactive agent aims to maximize the expected acceptance rate of the proposed tasks" (§3.1).

Liu et al. [42] is the strongest content-quality yardstick and brackets timing on purpose. Across 328 synthesized environments, six domains and 22 LLMs, it scores target planning (best: DeepSeek-R1, 7.60/10) and dialogue guidance (best: Claude-3.7-Sonnet, 9.01/10) with a human-validated judge (weighted κ 0.826 / 0.721). But every environment is pre-certified as warranting proactivity, and the trigger factor arrives as static text. There is no abstention and no cost of interrupting: false alarms are structurally unmeasurable.

Tang et al. [23] is the only benchmark on authentically real traces (28,528 events over 500+ hours of screen recordings) and writes cost into its metric semantics: "precision quantifies interruption cost (low precision causes alert fatigue), while recall measures need coverage" (§1). Performance on *when to assist* is near chance: best zero-shot 64.4% accuracy, with precision 51–61 against recall 81–99 across models, because models trigger constantly. Its sharpest finding indicts the rest of the roster: human inter-event times are heavy-tailed with burstiness 0.787, and "even with realistic candidate time points, the LLM does not naturally reproduce the bursty timing in human behavior" (§4.2). The blind spot is the target: ground truth is the moment the user actually opened an LLM, so an agent firing early enough to make that query unnecessary scores as a false positive.

Wang et al. [43] is the only work whose headline metric is timing itself, and it reports the loudest failure number in the literature. On 1,500 ten-second segments and 13,500 annotated pairs, intervention-timing accuracy is 12.50 (Phi-4), 13.36 (Gemini 1.5 Pro), 17.67 (Gemini 2.0 Flash) and 14.44 (Gemini 2.5 Pro), while the same models reach 51.65–56.08 macro-F1 on merely detecting that a social interaction is under way: "current OLLMs, while capable of detecting that a social interaction is occurring, still struggle to pinpoint the precise moment for intervention" (§5.2.1). It also revives the interruptibility state the deep-learning era dropped: "A positive SFD acts as a global veto, suppressing prompts when the wearer is cognitively busy" (§4.3). But the veto is boolean, not priced, and the ground truth is a mechanical disjunction of two cue labels (App. C.3), so "a user would least likely want to be disturbed" is asserted, not measured. The social cost is named; it is not charged.

Wang et al. [44] contributes the most sophisticated timing metric, and its blind spot is this survey's thesis in miniature. Within each ground-truth reply window, responses at $\tau_1 < \cdots < \tau_P$ are scored $s_p \in \{0, 1, 2\}$ by an LLM judge over the *accumulated* predictions; PAUC is the normalized area under the resulting step-polyline of cumulative correctness against time, anchored at $(t^{\text{start}}, 0.5)$ because "providing no response is preferable to giving entirely incorrect answers" (§3). It is human-validated: agreement with pairwise human preference rises from κ 0.23–0.31 at ω=1 (time-agnostic) to 0.30–0.45 at ω=0.5, against a human–human ceiling of only 0.34–0.55 (unweighted κ, matching the preceding figures). The authors concede that "all Cohen's kappa scores, including the agreement between human annotators, are relatively low" (§6.2). But responses *outside* every window go unscored, and redundant ones *inside* a window are free: MMDuet repeats itself on 99.4% of its egocentric turns and pays only in a diagnostic side table, while VideoLLM-Online's near-total muteness scores a respectable 25.0, the silence floor. PAUC prices lateness and wrongness. It does not price speaking. It is half a metric: a validated benefit curve $V(t)$ with no $C(t)$ beneath it.

Kundu et al. [45] is the most decision-theoretically mature entry and the field's first unification. Pro^2Bench re-annotates five public corpora plus a new smart-glasses set into 29,337 videos and 42,275 evaluation clips, each carrying a goal, a golden plan, and per-decision-point intervention labels (their Table 1, which we cite because counts do not reconcile across the paper). Its Out-of-Plan schema is the contribution: three deviation types (step omission, incorrect ordering, substitution/technique error), a human-marked onset ("the first frame at which the erroneous action becomes visually distinguishable from the correct action," $\pm 2\,\mathrm{s}$ tolerance), and a paired human-written recovery utterance. It is the first supervised task scoring both the catch and the correction, and its G-Mean F1 "collapses to 0 whenever either class is never predicted, directly penalizing the degenerate all-interrupt and all-silent policies that a raw accuracy metric would reward" (§3.3.1); no zero-shot baseline exceeds .51 average G-Mean F1 or .37 average PQS. Yet PQS assigns 0 to a false positive and 0 to a false negative: interrupting mid-pour with something irrelevant and silently watching the dish burn score identically. The release tags safety-critical steps precisely to enable "safety-aware interrupt suppression"; no metric uses the tags. It also stays script-anchored: deviations are experimenter-scripted with templates shared across splits, and §8 concedes "within-distribution generalization only."

Lepine et al. [46] is the empirical price tag, and it is not a benchmark. Observing 34 finance professionals across 1,176 turns and 2,352 utterances of real GPT-4o-assisted valuation work, it regresses expert-graded subtask quality on per-utterance cognitive load. Extraneous load carries $\beta = -0.472/-0.485$ ($p < 0.01$), about three times the coefficient on intrinsic task difficulty ($-0.147/-0.149$), and the decomposition names the culprit: "Among all behavioral indicators examined, unsolicited task switching by the model emerges as the strongest and most robust negative predictor of output quality" (§7.3), $\beta = -0.271$, the only disruptor surviving joint specification, while sprawl (0.064) and truncation (-0.051) go non-significant. Initiative, not volume, is what costs: identical structural complexity helps when the *user* introduces it ($+0.166$, $p < 0.001$) and does nothing when the *model* does (-0.025, n.s.). This is the C_{FA} that PRISM leaves as a swept hyperparameter (§4), measured in real professional work. The measurement is correlational, single-domain, N=34, with load indicators the authors themselves call "theory-grounded estimates rather than fully validated instruments" (§7.6). It measures the quantity a gate would need, and stops.

The roster confirms the field's own capability survey [2] while inverting its conclusion. That survey subordinates proactivity to procedural assistance and lists false-alarm rate among its metrics; what it never asks (and what none of the eight measurements above answers) is what an interruption *costs*, or what the user was doing when it arrived.

7.2 Feedback depth: the L0–L3 ladder

Benchmarks are offline proxies. A deployed assistant has a ladder of live signals with a structural property (Figure 3): each rung is a truer measure of intervention quality than the one below, and each is harder to attribute to the intervention that caused it.

L0: explicit response. Accept, dismiss, reject. ProactiveBench's annotation is built on it, and Liu et al. [22] measures it behaviorally: acceptance 0.676 under well-timed help versus 0.444 (random) and 0.388 (anti-timed), $H = 19.86$, $p < 0.001$. Cheap, instant, perfectly attributable, but biased. Gating on acceptance alone drives false alarms to 62.50%, "because users often accept helpful but ill-timed suggestions" [19, §4.3-D]. Acceptance measures whether the content was wanted, not whether the moment was right.

L1: implicit behavior. What the user does next. This is Pielot et al. [13]'s receptivity signal (voluntary consumption of optional content), and Liu et al. [20] sketches the desktop analogue (accepted suggestions mark fire events, dismissals mark over-fire events) without implementing it.

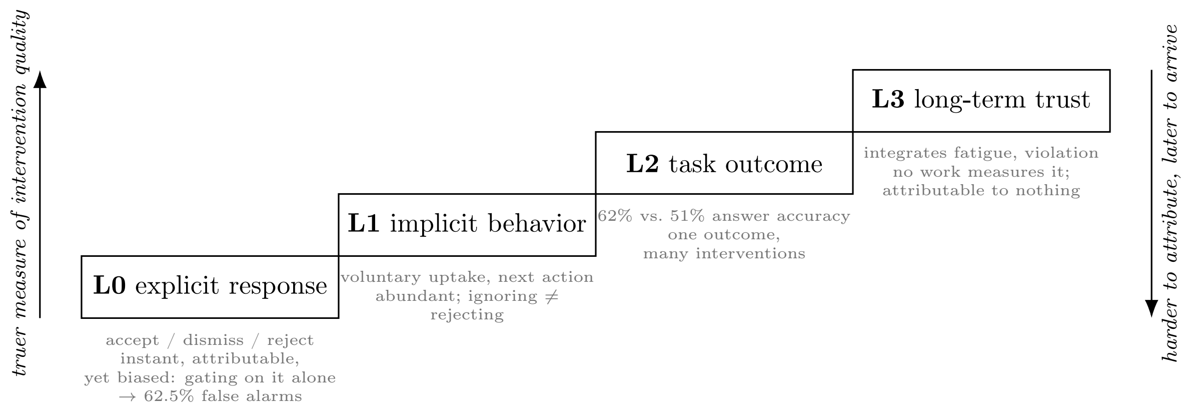

Figure 3: The L0–L3 feedback ladder. Each rung is a truer measure of what proactivity is for, and each is harder to credit to the intervention that caused it: an explicit dismissal arrives in seconds and gating on acceptance alone drives false alarms to 62.50% [19]; the only content-controlled task-outcome effect spans a twenty-item lab block [22]; and no work in this survey reports a longitudinal retention metric at all.

Abundant, weakly attributable: ignoring a suggestion is not rejecting it, and no system distinguishes the two.

L2: task outcome. Did the user do better work? Liu et al. [22] is the only content-controlled demonstration: identical model, sensing and help text, with only the threshold-update rule varying, yielding 62% answer accuracy under aligned timing versus 51% (anti-aligned) and 55% (random). These are honest condition effects of 7–10 points (LME -0.103, -0.070). Lepine et al. [46] is the professional-work analogue. This is the first rung measuring what proactivity is *for*, and attribution is already hard: a session holds many interventions and one outcome.

L3: long-term trust. Does the user keep the assistant on? Only this integrates alarm fatigue, social embarrassment, and the accumulated irritation of being right at the wrong moment. Liu et al. [22] catches its shadow in one lab session (benevolence -0.667, NASA-TLX effort $+0.562$ under mistimed help); Lu et al. [18] names it as the risk its 50% false-alarm rate incurs; Pielot et al. [13] shows its distributional form, buying precision by writing off $\sim$15% of users as never-opportune. No work in this survey reports a longitudinal disabling or retention metric. L3 is the truest reward and, at horizons of weeks, is attributable to no single intervention at all.

7.3 Self-improvement: three levels of maturity

Level 0: log only. The overwhelming majority: triggers fire, logs accumulate, and the threshold is a constant chosen once. Liu et al. [20] is explicit: $\tau = 0.5$, "the sigmoid decision boundary" (App. E.8), fixed across all fourteen backbones and never tuned.

Level 1: per-user threshold adaptation. One implemented instance exists. Liu et al. [22] runs a closed loop: initial $\theta = 12$, step $\delta = 1$, updated after each item by a six-branch rule over (offered? accepted? correct?) with a hand-chosen asymmetry (-4δ for a missed struggle, $+4\delta$ for a declined offer). It is a per-user bandit in all but name, and it identifies the causal channel: trigger *frequency* does not differ across conditions ($H = 2.23$, $p = .328$) while acceptance does. Well-timed help gets taken up, and uptake buys the accuracy. But the multipliers are the entire cost theory (never derived, never measured), and the system fires on $\sim$73% of trials at $\sim$33% specificity, tolerable only inside a twenty-item lab block.

Level 2: policy learning. Nothing online exists. Lu et al. [18] is the offline version, a fine-tuned reward model standing in for the user, and two pathologies follow. It is trained on the same pipeline's annotations that produce the labels it is scored against, so "acceptance" is circular; and when its feedback is fed back, the policy collapses rather than improves: GPT-4o's recall falls from 98.11 to 56.76 because "models stay silent once they receive feedback" (§4.3). Fu et al. [19] distills its gate from a teacher, keeping it auditable but equally offline, with no online recalibration (their §6). The field has an offline reward model a policy can overfit into muteness, and no deployed system that learns *when* from the user in front of it.

7.4 The asymmetry, and why reward ⊆ eval

The available signals are asymmetric, and the asymmetry determines what any RL objective built on them will optimize. Negative evidence is cheap, immediate, abundant: a dismissal arrives in seconds, false alarms are countable (22.94% for the best cost-sensitive gate, 50.22% for the strongest ProactiveBench agent), and irritation is legible in the next user action. Positive evidence is weak, diffuse, late: an intervention that saves ten minutes tomorrow produces no signal today. Worse, the strongest form of success is invisible by construction: ProAgentBench's positive label is the moment the user opened an LLM, so an agent that intervenes early enough to make that query unnecessary has, by the benchmark's own accounting, committed a false positive and erased the evidence of its own benefit. Optimize these signals as they stand and the gradient points toward silence, exactly the collapse Lu et al. [18] observed once its reward model entered the loop.

The contrast with the action layer is precise. Wang et al. [33] shows multi-turn RL scaling to long-horizon agentic behavior (OSWorld 47.5, AndroidWorld 73.3) because the reward has ground truth: the environment adjudicates completion. Intervention timing has no adjudicator, and the nearest substitute is PQS, which scores a false positive and a false negative identically at 0 [45]. That is not a noisy reward but a reward indifferent to the distinction the entire problem consists of.

Hence the containment. *Reward* ⊆ *eval* constrains what can be trained: an RL objective for proactivity is expressible only over quantities some evaluation scores, and today those are acceptance (biased), lateness (with no price on speaking), and class-balanced accuracy (pricing both error types the same). Each omits the term the 1999–2017 literature spent twenty years formalizing (§2), and each scores a policy that cannot be conditioned on the user it is scored against (§6). The gating investment is not a better policy or a bigger model. It is a benchmark that prices the interruption, and §8 takes up what that requires.

8 A Benchmark Design for Open-World Intervention Timing

This section is a design, not a result: an annotation protocol and a metric suite for intervention timing in open-world egocentric streams, with an explicit cost term. We publish the design before the data for the same reason the rest of this survey exists: the field's benchmarks each measure a slice of the decision (§7), and until the whole decision is measurable, neither training nor comparison can target it. We invite collaboration on the instantiation.

8.1 What existing benchmarks cannot answer

Consider the question a timing policy must answer at deployment: *of the moments you chose to act, how many were welcome; of the moments that needed you, how many did you catch; and how much did your mistakes cost?* No existing benchmark answers all three. ProactiveBench [18] labels proposals accept/reject but its events are desktop-synthetic and its one-sided use collapses agents into silence (§4.2). PAUC [44] is a genuine timing metric, human-validated, but it carries no cost term, so a policy can score well while interrupting badly. EgoSocial [43] makes timing the metric itself, but for the social slice only. HoloAssist's intervention-type prediction [41] presupposes an intervention is happening and asks only which kind. Pro^{2}Bench [45] unifies five procedural benchmarks yet remains script-anchored: "off-plan" is still defined against a known plan. The open-world case, with no script, real streams, and priced errors, is uncovered.

8.2 Annotation protocol

Substrate. Egocentric video segments drawn from existing corpora (HoloAssist for collaborative manipulation, Ego4D for daily life [41, 47]), re-annotated. HoloAssist is the natural seed: its recorded instructor utterances are ground-truth *human* interventions, with timing, made by a party who could see the same stream the model sees.

Labels. Annotators mark decision *windows*, not instants, because interventions are rarely uniquely timed. Each window carries a three-way label: MUST (a mistake or hazard is unfolding; a competent assistant intervenes), MAY (an intervention would plausibly be accepted; silence is also acceptable), and MUST-NOT (intervention would be a violation: the user is concentrating, conversing, or the matter is trivial). The three-way scheme is the point. Binary schemes force the MAY mass into one side and thereby erase the region where cost-benefit reasoning, rather than classification, does the work; a policy's behavior *inside the* MAY *band* is where conservatism, personalization, and θ become visible.

Disagreement is signal. Receptivity is personal (§6); annotators will disagree, and the protocol records rather than adjudicates. Each window keeps its label distribution plus a minimal annotator profile, enabling evaluation both against the majority and against persona-conditioned subsets. We budget deliberately small: one scenario family, 200–300 windows, three annotators each. That is the scale one group can produce with rigor, following the precedent that ProactiveBench's reward-model subset needed only 1,640 training events to be useful [18].

8.3 Metric suite

Four numbers, reported together and never collapsed into one:

1. **Timing recall**: fraction of MUST windows in which the policy acted. The benefit side.

2. **Violation rate**: fraction of the policy's actions falling in MUST-NOT windows. The cost side, kept separate so that, unlike F1, silence cannot buy safety invisibly: a policy that never speaks has perfect violation rate and zero recall, and both numbers are visible.

3. **Cost-weighted score**: $\sum_i b_i - \lambda \sum_j c_j$ over acted windows, with per-window costs c_j scaled by label (MUST-NOT $\gg$ unlabeled) and λ *reported as a curve, not chosen*: policies are compared by dominance across a λ range, which is precisely the exchange-rate question Pielot et al. [13] declined and Fu et al. [19] fixed per deployment. The benchmark makes it a swept axis instead of a hidden constant.

4. **Graduated-action credit**: where the policy supports low-cost partial interventions (a chime, a glance-able cue, a backchannel; §5), violations by a graduated action are discounted relative to full interruptions, following the cost menu the full-duplex literature already measures at millisecond scale [36].

Two reporting rules complete the suite. Per-user (or per-annotator-persona) variance is reported alongside every aggregate, because the strongest engagement model of the mobile era achieved its precision partly by writing off ∼15% of users [13], a failure mode invisible in aggregates. And silence on MAY windows is never penalized: the benchmark must not repeat the one-sided-penalty mistake that ProactiveBench's own ablation exposed (§4.2).

8.4 What this design does not solve

Offline replay cannot measure the counterfactual benefit of an intervention that changes the user's subsequent behavior, nor delayed benefit at all (§9); annotated receptivity is a proxy for the real thing, one level better than click-through but below deployment A/B; and 200–300 windows will not train a policy, only rank several. These are accepted costs of a design whose purpose is to make $\mathbb{E}[\text{benefit}] - \mathbb{E}[\text{cost}]$ *visible* for the first time in the open world. The v1 instantiation calls for on-device capture, real interventions, and L1 implicit feedback (§7). It requires hardware in daily use, and it is where this design is headed.

9 Open Challenges

Six problems remain open. They are not independent: the first is upstream of the others, in the sense that a solution to any of them would still be unverifiable without it. For each we state what is blocked, the strongest partial answer, and the most plausible first step.

1. Reward without ground truth. GUI agents train under RL because the environment adjudicates completion [33]. Intervention timing has no adjudicator, and every surrogate breaks in a diagnosable way. Acceptance is biased: gating on it alone drives false alarms to 62.50%, "because users often accept helpful but ill-timed suggestions" [19, §4.3-D]. Class-balanced accuracy is indifferent to the distinction that matters: PQS scores a false positive and a false negative identically at 0 [45]. Value-accrual metrics price lateness but not loudness [44]. And the one benchmark built on real traces defines its positive label as the moment the user opened an assistant [23], so an agent that intervenes early enough to make the query unnecessary is scored a false positive, its success erased from the record. Blocked downstream: any RL objective, any online policy learning, any principled model selection.

The first step is a label, not an algorithm: a *three-way per-moment human judgment* (should intervene / could intervene / must not intervene) on streams that already carry the surrounding annotations, seeded from HoloAssist's existing conversational-intervention labels (§8). A binary label cannot separate a moment where silence is merely suboptimal from one where speech is a violation, and that separation is what turns the cost term from a hyperparameter into a measurable quantity.

2. The cost term: modeled in fragments, unified nowhere. The honest 2026 status is no longer "nobody models interruption cost" but that four communities each model one component and none cites the others. Fu et al. [19] prices *false alarms*, deriving its gate (intervene iff $p_{\text{accept}} \geq \tau(p_{\text{need}}) = C_{\text{FA}}/(C_{\text{FA}} + p_{\text{need}} \cdot C_{\text{FN}})$) from Bayes-risk minimization and cutting the strongest prior agent's false-alarm rate on ProactiveBench from 50.22% to 22.94% (their Table 1); but C_{FA} and C_{FN} are free knobs swept from 1:4 to 1.2:1, never measured from a user, and its §6 concedes that LLM-judge proxies "cannot fully capture the cognitive load of ill-timed interruptions during complex flow states." Lepine et al. [46] measures exactly that: extraneous load at $\beta \approx -0.48$ on graded work quality, roughly three times the weight of intrinsic difficulty, with model-initiated task switching the strongest single disruptor at $\beta = -0.271$. It attaches that measurement to no decision rule. Wang et al. [43] models the *social* component as an absolute veto ("A positive SFD acts as a global veto, suppressing prompts when the wearer is cognitively busy," §4.3) rather than a price a sufficiently urgent message could outbid. The gating architectures price *compute*: Liu et al. [20] replaces an always-on LLM trigger with a 1.16M-parameter graph head at 11.13 ms and ∼220 MiB, and PRISM routes only ∼11% of cases to slow reasoning. Four costs, four literatures, no unified θ. None of them *reuses* the formalisms of Horvitz et al. [5], Horvitz and Apacible [7], or Fogarty et al. [9], who between them built the expectation-over-attentional-state cost, the dollar-denominated learned cost estimator, and the shallow-sensor interruptibility predictor these fragments are independently rediscovering. Where the interruption literature is acknowledged at all, it is cited and not modeled: Liu et al. [20] cites Iqbal and Horvitz's notification work in its related-work section and then prices only compute.

The first step is a substitution rather than a synthesis: take PRISM's threshold, which already has the right shape, and replace its swept C_{FA} with a load estimate of the kind Lepine et al. [46] shows is computable from the interaction trace. A measured cost inside a derived threshold closes the loop the lineage left open, and it is testable on data that already exists. Making C_{FN} time-varying (rising toward a deadline, as PRISM's §6 proposes and does not implement) is the natural second move.

3. Script-free timing. Deviation-from-script is the field's workhorse trigger, and it works because a script encodes *why* an intervention is worth its cost: the user is about to do something that will not achieve their goal. Open-world daily activity has no script. Kundu et al. [45] gives the first serious answer, and it is worth being exact about what it does: it keeps the script but makes *leaving* it recoverable, annotating three deviation types, a human-marked onset with ± 2 s tolerance, and a paired human-written recovery utterance. But the deviations are experimenter-scripted, the test split shares templates with training, and §8 concedes "within-distribution generalization only." More tellingly, plan conditioning *hurts* the baselines. Claude's out-of-plan detection collapses from 72.1% to 28.4% with a predicted plan, "because a plan biases the model toward silence exactly when it should flag a problem" (§4.1), and oracle plans underperform zero-shot plans in 63% of testable pairs, ruling out plan accuracy as the bottleneck and pointing at plan *utilization*. A recoverable script is the best available answer, not the final one.

The first step is to learn the trigger's latent structure instead of authoring it. Liu et al. [20] shows that "when to wake" is temporal-graph node classification rather than a language problem, and that the structure the OS already keeps (entities, recurrence, inter-event gaps) carries the signal. Ported to open-world egocentric streams, the question becomes whether an unsupervised model of a user's habitual activity graph can supply the counterfactual ("this is not what you usually do next") that a script currently hard-codes.

4. Credit assignment for delayed benefit. An intervention whose value materializes tomorrow generates no signal today (§7.2). The strongest demonstration that timing carries outcome effects at all spans a single twenty-item block [22]; the strongest evidence from real professional work is correlational [46]. Nothing in the literature attributes a downstream outcome to a specific intervention across a session boundary, let alone across days. This is what makes L3 trust the truest and least usable signal: by the time a user disables the assistant, the intervention that cost them is months gone.

The first step is to import a formalism the field already has and has not used. Horvitz et al. [5] computes the expected cost of *delayed* action against a model of when the user would have encountered the information anyway: an explicit counterfactual baseline for benefit. Instantiating that baseline ("what would this user have done in the next N minutes without me?") yields a per-intervention credit signal that does not require waiting for the outcome, and it is estimable from precisely the behavioral traces Tang et al. [23] already collects.

5. Personalized thresholds. One θ is wrong across users, contexts, and times of day, and this is the oldest and most consistent result in the lineage. Pejović and Musolesi [12] showed it in the wild a decade ago, when an online per-user classifier beat the population model trained on everyone else, and showed the threshold is not even stable *within* a user: sentiment toward an interruption degrades with the interruption load of the preceding two hours, so the cost side is path-dependent, not a constant to be found. Pielot et al. [13] found that four features summarizing a user's own past responses outperform all 197 context features combined, for a five-fold precision lift over base rate. Liu et al. [22] implements the per-user loop and shows it changes uptake (acceptance 0.676 versus 0.388 anti-timed) rather than frequency ($H = 2.23$, $p = .328$, n.s.). And Horvitz and Apacible [7] supplies the warning that should govern every attempt: cost models transferred across users scored .28/.32 accuracy, *below* majority-class guessing, against .73/.64 within-user (their Tables 1 and 3). A population-level threshold is not weak personalization; on the only transfer evidence available, it is worse than a constant.

The first step is to make the personal component the *only* learned component. Freeze a population model of *what* help is relevant, which current systems already learn well, and learn per-user only the scalar deciding *whether* to say it, initialized from the population prior and updated from L0/L1 feedback. That scalar is the smallest object personalizable from the handful of interactions a new user supplies, and it is the one thing Horvitz and Apacible [7] shows cannot be transferred.

6. Benchmark fragmentation. Each benchmark measures a slice and the slices do not compose (§7.1): HoloAssist scores intervention *type*, conditioned on the intervention occurring; ProactiveBench scores accept/reject with no temporal metric at all; ProactiveEval scores dialogue quality in environments pre-certified as warranting speech; EgoSocial scores timing, but only over the social slice at ten-second granularity; PAUC scores the benefit curve and prices no interruption. Pro^{2}Bench is the first serious unification: five public corpora plus a new smart-glasses set, normalized to 29,337 videos and 42,275 evaluation clips with per-decision-point intervention labels. It nonetheless inherits the script anchor from all five, since deviation-from-plan is what its labels are defined against. Unifying five script-anchored benchmarks yields a large script-anchored benchmark. No benchmark covers open-world intervention timing with an explicit cost term, so no two systems in this survey can be compared on the quantity the survey is about.

The first step here is additive, not architectural. Pro^{2}Bench already tags safety-critical steps in its release, explicitly to enable "safety-aware interrupt suppression," and no metric uses those tags. A cost-weighted PQS, in which a false positive during a tagged step costs more than one during idle time and a missed hazard costs more than a missed cosmetic error, requires no new data collection and would make the asymmetry the field keeps naming finally visible as a number. That is the minimum viable cost term, and it ships on data that already exists.

10 Conclusion

This survey traced a mismatch of two halves. Between 1999 and 2017 the interruptibility literature formalized the cost of speaking with real rigor: an expected cost of alerting computed over an inferred attentional state, a learned interruption price denominated in dollars, sensor estimators that beat human observers at judging when not to interrupt. What it had no access to was an actor capable of delivering benefit worth the machinery (§2). The 2024–2026 proactive wave inherited the opposite half. It has actors: agents completing long-horizon tasks against verifiable goals, and egocentric perception that reads intent from a live stream (§3, §5). It rebuilt the decision layer without the inheritance, and is now rediscovering the cost term one component at a time.

The fragments are converging. Three independent 2026 systems arrived at the same cheap-gate, expensive-actor architecture (§4). Expected utility has been rebuilt inside an LLM agent, with a Bayes-derived threshold over calibrated need and acceptance probabilities. Timing has become the scored quantity rather than a side condition, and the first numbers the field got back sit between 12.50% and 17.67% across four frontier models [43, §7]. Cognitive load has been measured in real professional work, at roughly three times the weight of intrinsic task difficulty [46]. Every piece of the synthesis exists, and nobody has assembled it: the four cost fragments (false alarm, cognitive load, social violation, compute) still sit in four literatures that do not build on one another, and that reuse none of the formalisms they are reconstructing.

What blocks the assembly is measurement, not modeling capacity. Reward is contained in evaluation: an objective for proactivity can only be written over quantities some benchmark scores, and the most sophisticated benchmark available assigns a false positive and a false negative the same score of zero [45]. Under that objective, interrupting a user mid-pour with something irrelevant and silently watching the dish burn are the same outcome. No amount of policy learning repairs a reward indifferent to the distinction the problem consists of.

The field's next concrete step is therefore a labeling decision. Collect the three-way per-moment judgment (should intervene, could intervene, must not intervene) on open-world egocentric streams, and attach a cost weight to the third class. It is the smallest artifact that makes intervention timing trainable, it builds on corpora that already exist, and Pro^{2}Bench's unused safety tags are the cheapest place to begin. The gating investment is the eval.

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